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Unlocking Potential: Increasing Stimulation Effectiveness Using Holistic Fracturing Diagnostics & Conductivity Enhancement-A Case Study From South-East Kuwait Field

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Abstract

The Ahmadi field located in South-East Kuwait has a prolific low matrix permeability carbonate formation with moderate to high oil viscosity. During the initial field development phase, the field started suffering from significant production decline and rapid reservoir pressure depletion. These challenges led to stimulating these wells, further driven by multiple factors—the desire to connect the natural fractures, to fully drain formation and to increase the Estimated ultimate recovery (EUR) of the well through achieving additional stimulated reservoir. This paper presents the entire process of identification of a potential candidate including operational preparations, execution, and analysis of post-treatment results.

After the well candidate selection, developing a solution phase that will accomplish the treatment's goal and addressing reservoir challenges such as stimulation of depletion, high fluid leak-off tendency is one of the most important challenges. The well was treated using High quality Nitrogen (N₂) foam with High Viscosity Friction Reducer (HVFR) PAD fluid and modified single phase retarded acid system (mSPRAS) to ensure deep and branched ramifications. The Nitrogen (N₂) Foam acid selected, is a finely dispersed mixture of nitrogen gas bubbles within hydrochloric acid liquid which increases the volume of the active acid, improves penetration, and diverts fluid from high permeability zones into low permeability zones. Holistic design diagnostics with memory gauge data, Diagnostic fracture injection test (DFIT) analysis and swift temperature log analysis for frac height computation was conducted, to ensure revalidation of fracture simulations and to tune the "Tailor-Fit" fracturing design rather than conventional cookie cutter method. Energized acidfracs helped in reducing fluid usage and promoting higher length growth with improved fracture width relative to low viscosity fluids- resulting in an optimum fracture geometry and superior conductivity enhancement.

Based on the comparative valuation completed on nearby wells, the use of energized fracs is shown to create significantly improved performance over those wells fractured with non-energized fluids. Temperature logs, bottom-hole data frac analysis, sonic log data confirmed that the fracture remains within this layer with an average half-length of 370 ft. The presence of existing natural fractures was inferred through a significant permeability increase observed, reaching up to 1.2 Darcy, likely attributable to

substantial fracture apertures. Analysis of diagnostic ISIP, closure pressure, and temporal changes indicated a well-connected fracture network extending away from the wellbore. Post treatment production increased 6 folds per day.

This paper summarizes the candidate selection criteria, application of this unique stimulation design in Kuwait, operational procedures, diagnostics tools, well clean-up and production performance aspects of this treatment. Success of this technique is crucial for the Ahmadi field to overcome production decline and reservoir pressure depletion and effectively stimulate this prolific carbonate formation, increasing the stimulated reservoir volume to continue rising field production capacity. The learnings presented can be utilized for similar challenges in other fields.

Introduction

The Mauddud reservoir in the Ahmadi Field of Kuwait (Figure 1) is a complex, heterogeneous carbonate system characterized by moderate petrophysical quality and challenging geomechanical behavior. It consists of a layered sequence of grainstones, packstones, and wackestones, with the MaD and MaE units serving as the most prolific zones due to their relatively high porosity (13–17%) and moderate permeability (<10 mD). The overall reservoir porosity can reach up to 31% in some regions, though significant heterogeneity exists due to interbedded clastics and variable shale content. Water saturation varies widely across layers, and the reservoir primarily operates under a solution-gas drive mechanism with limited natural aquifer support. Geomechanically, the Mauddud formation displays tight rock characteristics with low natural injectivity, requiring hydraulic fracturing to enhance flow. Multi-well interference tests and pressure transient analysis reveal high vertical connectivity and significant anisotropy, with faults acting as both conduits and barriers to flow. This reservoir is composed of tight complex carbonates with net gross pay that ranges from 10 up to 25 feet. Its porosity response is generally low, as well as its matrix permeability (Alzankawi, O., et.al. 2014).

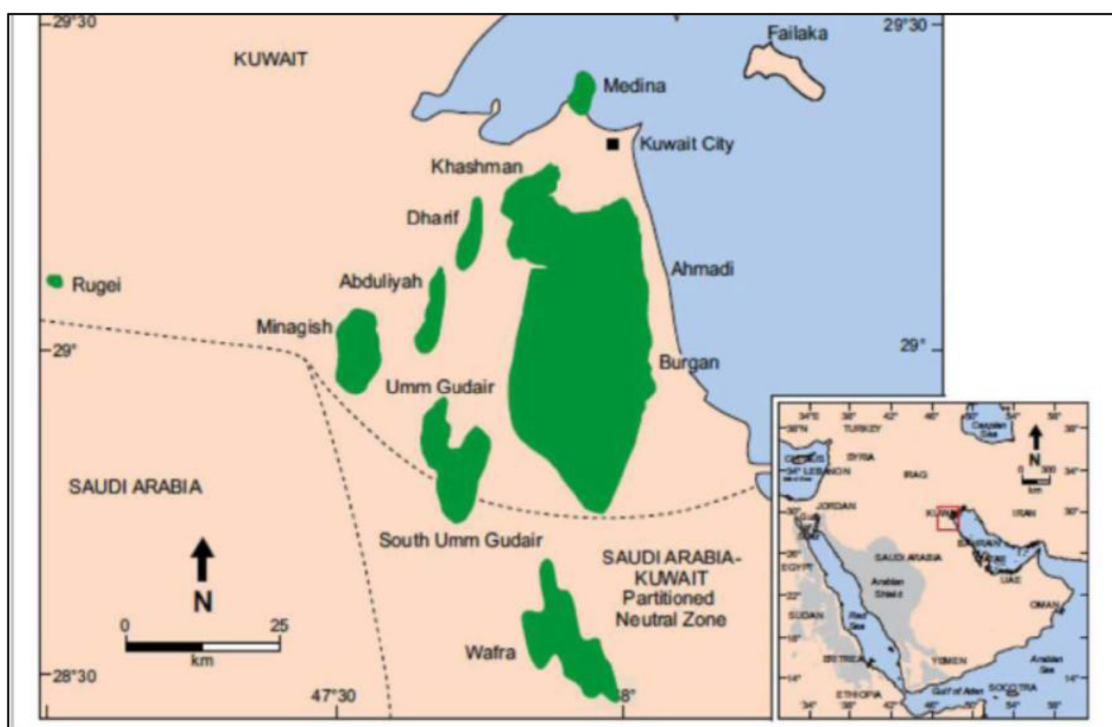


Figure 1—Location of Ahmadi Field- Southeast Kuwait.

Mauddud reservoir zones vary significantly due to heterogeneity; thus, geomechanical logging (e.g., sonic logs, density logs) is used to map these properties across the field. Brittleness is rock's tendency to

break or fracture rather than deform plastically. In hydraulic fracturing, more brittle rocks are preferred because they tend to fracture more easily and create complex, productive fracture networks. In Mauddud, zones with lower shale volume and higher carbonate content are typically more brittle and thus better candidates for fracturing. Young's modulus quantifies a rock's stiffness – that is, its ability to resist deformation under stress and is defined as the ratio of stress to strain in the linear elastic region. A higher Young's modulus indicates a stiffer rock that can store more elastic energy and generate wider fractures when stimulated. In the Mauddud reservoir, typical values of Young's modulus may range between 3 to 4 million psi, depending on lithology and porosity. Higher values indicate more competent limestone or tight dolomite zones. Poisson's ratio describes the lateral strain to axial strain ratio in a material when compressed. Lower Poisson's ratio values (e.g., 0.15–0.25) typically correlate with brittle and fracture-prone rocks. In Mauddud, layers with lower Poisson's ratio and higher Young's modulus are the most favorable for fracturing operations (Figure 2).

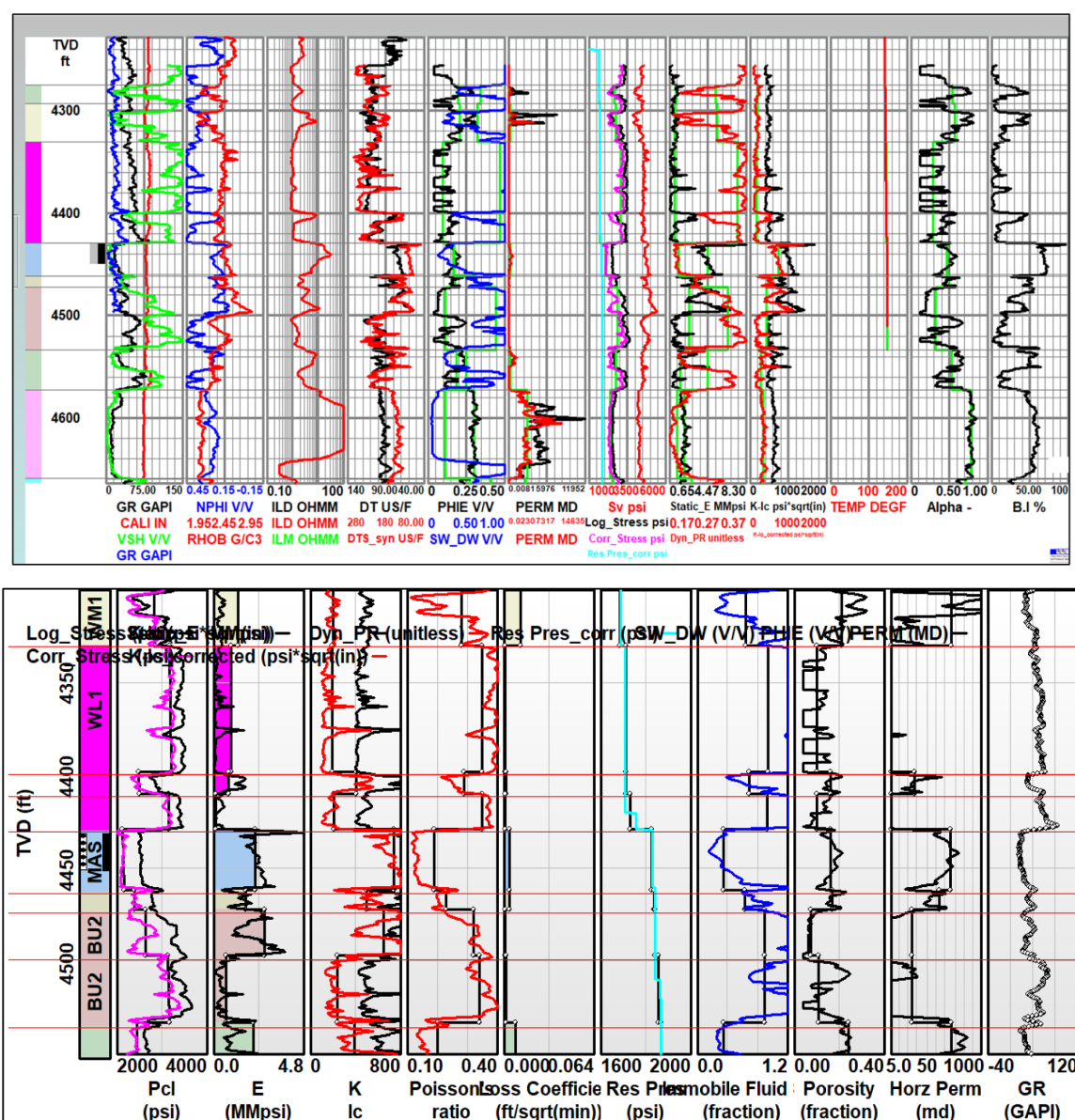


Figure 2—Well A Petrophysical and Geomechanical Properties

Completion Strategy

An essential component of a successful stimulation treatment is the careful selection of candidate wells. Multiple factors must be considered when evaluating the suitability of a well for stimulation, and without a thorough understanding of the treatment objectives, it becomes challenging to identify wells with the highest potential for economic return. The primary objective of the stimulation treatment is to enhance the Estimated Ultimate Recovery (EUR) by increasing the stimulated reservoir volume along the lateral section.

Well-A, located in the Ahmadi field, targets the Mauddud formation, which is considered to have significant potential. The reservoir pressure is approximately 0.4 psi/ft. This well was drilled and completed using a 7-inch casing, with specific intervals selected for stimulation, as illustrated in [Figure 3](#). The formation exhibits a vertical thickness of approximately 30 feet in this horizontal section. Key reservoir and well parameters are summarized in [Table 1](#). Based on a detailed evaluation, Well-A was selected as a suitable candidate for hydraulic fracturing.

Table 1—Well A Properties

Parameters	Value	UOM
Reservoir Pressure	1780	psi
Reservoir Temperature	148	oF
Target Zone	Mauddud	-
Zone of Interest	4430 – 4460	ft (MD)
Perforations	4430 – 4450	ft (MD)
Perf Phasing	36	o
Shots Density	10	spf

Energized Acid Fracturing (EAF) Strategy

In fracture acidizing, acid is pumped into created fractures at pressures typically higher than the formation's fracture pressure. This stimulation technique either enlarges existing flow channels or creates new ones, enhancing the movement of oil or gas from the reservoir to the wellbore. The effectiveness of acid penetration in this process depends on several factors, including the acid's flow rate, its reaction rate with the formation, the surface area of contact between the acid and the fractures, and the rate at which the acid leaks off into the formation. The maximum acid penetration occurs when the initial volume of injected acid is fully neutralized. Although subsequent volumes of live acid may further etch the fracture surfaces, they typically do not extend the penetration distance beyond that achieved by the first increment. This etching creates an irregular fracture surface, which helps prevent complete fracture closure when the pressure is released.

An added benefit to production comes from the creation of a zone with increased matrix permeability near the fracture surfaces, due to acid leak-off into the surrounding formation. This enhanced permeability can improve well productivity even if the fractures nearly close after treatment. In naturally fractured formations, the velocity of acid flow is primarily determined by the injection rate. Maximum penetration occurs when the injection rate generates a pressure just below the threshold required to form new fractures. Exceeding this pressure tends to widen existing fractures or create new ones, which reduces acid velocity and, consequently, its penetration. To achieve optimal flow capacity, the acid must etch an uneven surface pattern on the fracture faces. While fractures may partially close after treatment, these irregular etchings help maintain connectivity between the wellbore and the deeper parts of the formation. Acid leak-off also contributes to creating a high-permeability zone around the fractures, supporting improved flow. Three primary factors that affect the extent and nature of fracture face etching are:

- Rock properties
- Type of acid used
- Contact time between the acid and formation

Developing an effective treatment design that meets the stimulation objectives while mitigating reservoir-specific challenges—such as stimulation under depletion and elevated fluid leak-off rates—represents a critical technical challenge. In this operation, the well was stimulated using high-quality nitrogen (N₂) foam in conjunction with a High Viscosity Friction Reducer (HVFR) as the PAD fluid, followed by the application of a modified Single-Phase Retarded Acid System (mSPRAS) to enable deep matrix penetration and complex ramification. The selected N₂ foam acid system, comprising a finely dispersed dispersion of nitrogen gas microbubbles within a hydrochloric acid base fluid, enhances the apparent acid volume, improves acid propagation efficiency, and facilitates effective diversion from high-permeability zones to less permeable reservoir intervals, thereby optimizing contact with unstimulated matrix regions.

The Single-phase retarded acid systems (mSPRAS) provide higher performance and operational efficiency benefits than systems based on unmodified HCl and emulsified acid. When combined with diverting fluid systems and modeling, they have proven to resolve many challenges with tight carbonate acidfracs. The new single-phase retarded acid has a low viscosity, close to the viscosity of unmodified HCl, which should result in low friction pressure and high carbonate dissolving power (Danican S. et.al, 2024). A comparative analysis of acid retarders on HCl reactivity has shown bench top reaction studies. Results from lab study indicate that the single-phase retarded acid system requires lesser pore volumes to breakthrough than the conventional acids over wide range of injection rates (Qamruzzaman M., et.al, 2023).

In acid fracturing operations, foamed acid systems typically comprise 40–80% nitrogen gas by volume and 60–20% aqueous acid phase, which is stabilized with surfactants and may also include gelling agents and fluid loss control additives. These systems are particularly advantageous in treating low-permeability carbonate reservoirs and formations containing water-sensitive clays, where conventional acid systems may

cause excessive fluid leakoff or formation damage. The entrained nitrogen provides an inherent energizing effect, contributing to elevated surface treating pressures and enhanced acid penetration into the formation. Additionally, the low hydrostatic head of foamed acid minimizes fluid loss, reducing the risk of damaging sensitive formation minerals and facilitating rapid post-treatment clean-up. Energized acidfracs helped in reducing fluid usage and promoting higher length growth with improved fracture width relative to low viscosity fluids- resulting in an optimum fracture geometry and superior conductivity enhancement.

In this diversion methodology applied during acid fracturing operations, the initial stage introduction of an aqueous solution containing a foam-generating surfactant, which is displaced into the formation using nitrogen gas. A portion of the surfactant adheres to the formation surfaces, both within fractures and in the surrounding rock matrix. Subsequent injection of nitrogen interacts with the retained surfactant, initiating in-situ foam generation due to shear and agitation effects. As foam formation progresses, its mobility decreases significantly owing to its high apparent viscosity and multiphase nature. This increased resistance to flow elevates the injection pressure until it exceeds the breakdown pressure of a new, untreated segment of the formation. At this point, flow is diverted from the previously stimulated interval, allowing acid to enter and fracture a new section of the reservoir. This staged process is repeated to ensure acid placement throughout the targeted treatment interval. The foam is pre-generated at the surface by injecting nitrogen into the surfactant-laden aqueous fluid prior to entering the wellbore. The pre-formed foam then follows the acid pad into the formation, where the gas-liquid multiphase flow results in increased flow resistance and elevated bottomhole pressure. This pressure buildup promotes the breakdown of additional formation segments, facilitating sequential acid fracturing along the interval.

The injection rate plays a pivotal role in governing the placement and propagation of both the acid and the foam diverter within the fracture network. An optimized injection rate is essential to ensure sufficient fracture extension, effective foam generation, and controlled diversion. Excessively low rates may lead to inadequate fracture propagation and insufficient foam development, whereas excessively high rates can result in premature foam breakdown or ineffective diversion due to turbulent flow conditions. Therefore, the injection rate is carefully engineered to balance acid fracture propagation with in-situ foam stability and diversion efficiency, maximizing reservoir contact and overall treatment success.

Well A Strategy

DFIT (Diagnostic Fracture Injection Testing)

The diagnostic injection test was executed to characterize fracture leakoff behavior, quantify the closure stress gradient, and assess fluid efficiency—parameters essential for optimizing the main hydraulic fracturing (mainfrac) treatment design. It is used to measure reservoir properties including stress, pressure and permeability, all of which are important parameters affecting fracture stimulation effectiveness and well performance (Liu, X., et.al, 2024). The test employed base fluid at a relatively low volume but was performed at a high injection rate to ensure dynamic conditions. The fluid volume was strategically selected to achieve sufficient fracture coverage across the intended pay zone. Pressure decline was continuously monitored during the shut-in phase (Figure 4) and analyzed until the transition to the pseudo-radial flow regime, enabling extraction of reservoir transmissibility and storage characteristics. G-function analysis was utilized to interpret the pressure falloff response and delineate the dominant leakoff regime. The analysis indicated a normal leakoff mechanism prevailing within the formation, characterized by a fracture gradient of 0.54 psi/ft, a closure gradient of 0.5 psi/ft, and a reservoir pressure gradient of 0.43 psi/ft (Figures 5 and 6). These results underscore the necessity of maintaining precise injection rate control to promote optimal fracture geometry and maximize half-length extension during mainfrac operations.

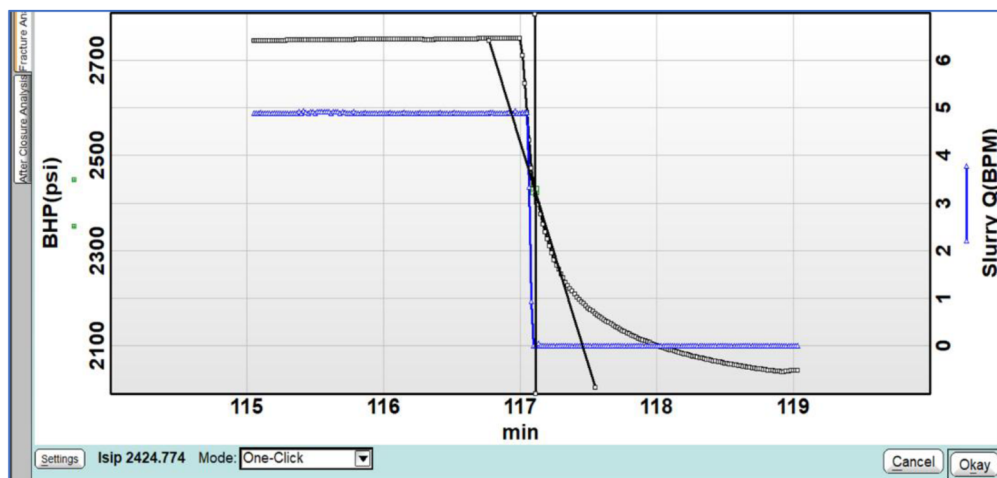


Figure 4—DFIT Analysis- ISIP Plot

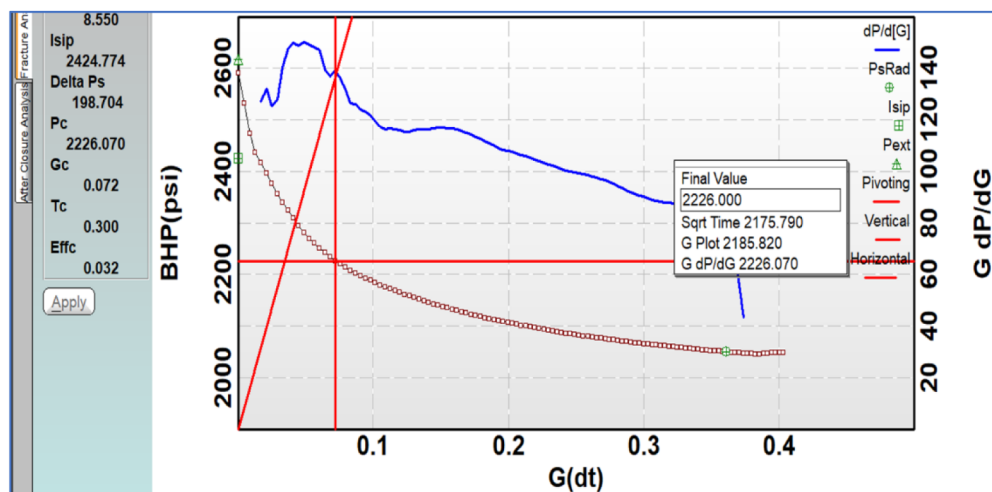


Figure 5—DFIT Analysis- G Function Plot

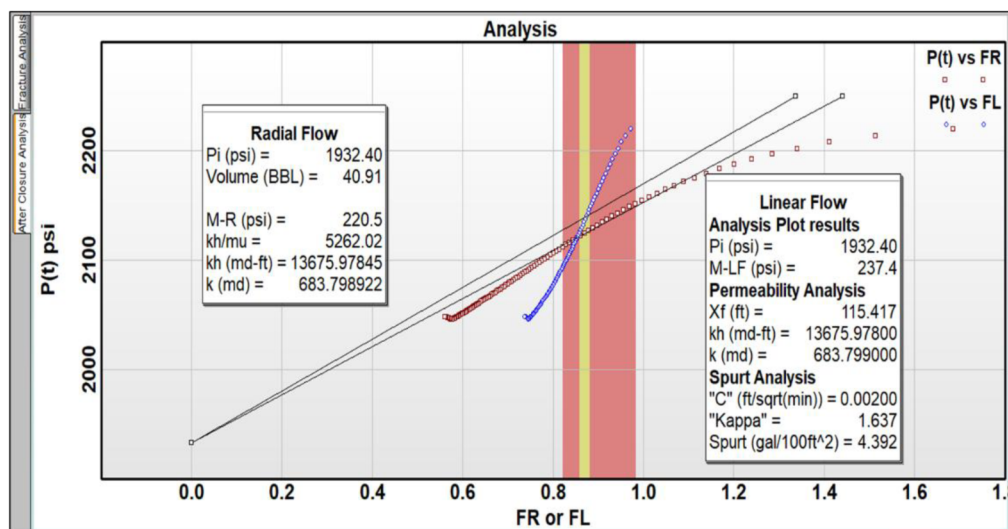


Figure 6—DFIT Analysis- After Closure Plot

Step-rate test (SRT)

The test was conducted to more accurately determine the fracture closure pressure and to evaluate both the extension rate and pressure required for fracture propagation. A step-rate test was implemented using short, successive pumping stages lasting 1–2 minutes or less, with progressively increasing pressure. In the case of a smooth, planar fracture that opens in direct proportion to fluid pressure, the net fracture pressure—defined as the difference between bottomhole treating pressure and closure pressure—is correlated with the injection rate (Figure 7). For this well, an extension rate exceeding 1.15 bpm and a bottomhole extension pressure of 2618 psi were calculated (Figure 8).

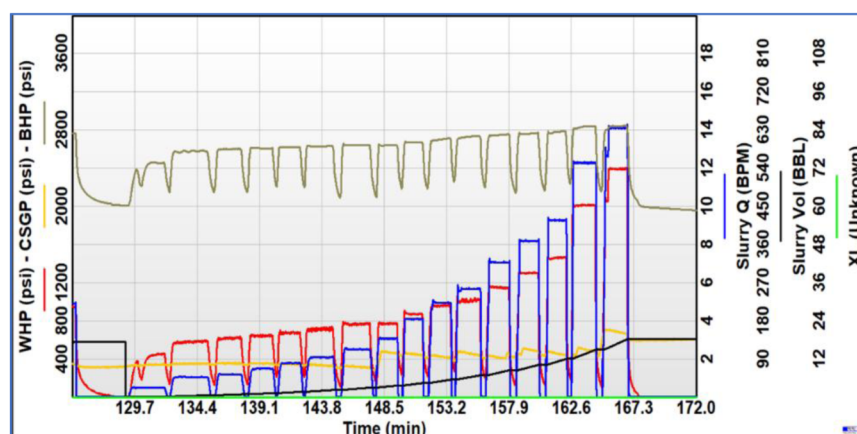


Figure 7—SRT Analysis- Pressure Plot

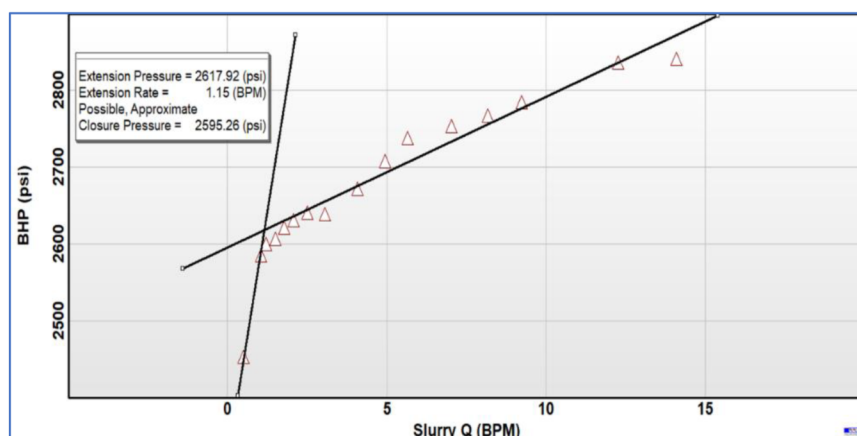


Figure 8—SRT Analysis- Extension Plot

MainFrac

Based on the comprehensive analysis of diagnostic testing and treatment evaluation, the MainFrac design was further optimized to enhance fracture conductivity and stimulation effectiveness. Key treatment parameters—including pumping rate, fluid volumes, and acid concentrations—were adjusted to improve conductivity and to counteract the high leakoff behavior observed in the formation (Table 2, Figure 9). A consistent high pumping rate was maintained to sustain net pressure and promote fracture propagation. Advanced fracture modeling and simulation techniques were utilized to predict fracture geometry, growth behavior, and fluid injection, enabling data-driven optimization of fracture design, and fluid systems for improved reservoir contact and production. Post-DFIT analysis, the simulation model (Figure 10) was recalibrated with the pressure response to ensure predictive accuracy. The final MainFrac design was refined

to achieve the desired fracture half-length and width, thereby maximizing wormhole creation and overall fracture conductivity (Table 3).

Table 2—MainFrac Schedule

Step Number	Step Name	BH Slurry Rate (bpm)	Surf Slurry Rate (bpm)	N2 Rate (scf/min)	B.H Foam Quality Check (%)	Slurry Vol. (bbl)	Total Slurry (bbl)	N2 Vol (gal)	Total N2 Vol (gal)	Pump Time (min)	Cum Pump Time (min)
1	PRE-FL	14	14.0	0	0%	50	50	0	0	3.6	3.6
2	PAD-1	14	7.7	3651	45%	300	350	1528	1528	39.0	42.5
3	ACID-1	14	7.7	3651	45%	100	450	509	2037	13.0	55.5
4	DIV-1	14	7.7	3651	45%	50	500	255	2291	6.5	62.0
5	PAD-2	14	7.7	3651	45%	150	650	764	3055	19.5	81.5
6	ACID-2	14	7.7	3651	45%	100	750	509	3564	13.0	94.5
7	DIV-2	14	7.7	3651	45%	50	800	255	3819	6.5	101.0
8	PAD-3	14	7.7	3651	45%	150	950	764	4583	19.5	120.5
9	ACID-3	14	7.7	3651	45%	150	1100	764	5346	19.5	139.9
10	POST-FL	14	7.7	3651	45%	50	1150	255	5601	6.5	146.4
11	CFA	14	7.7	3651	45%	50	1200	255	5855	6.5	152.9
12	FLUSH	14	14.0	0	0%	70	1270	0	5855	5.0	157.9

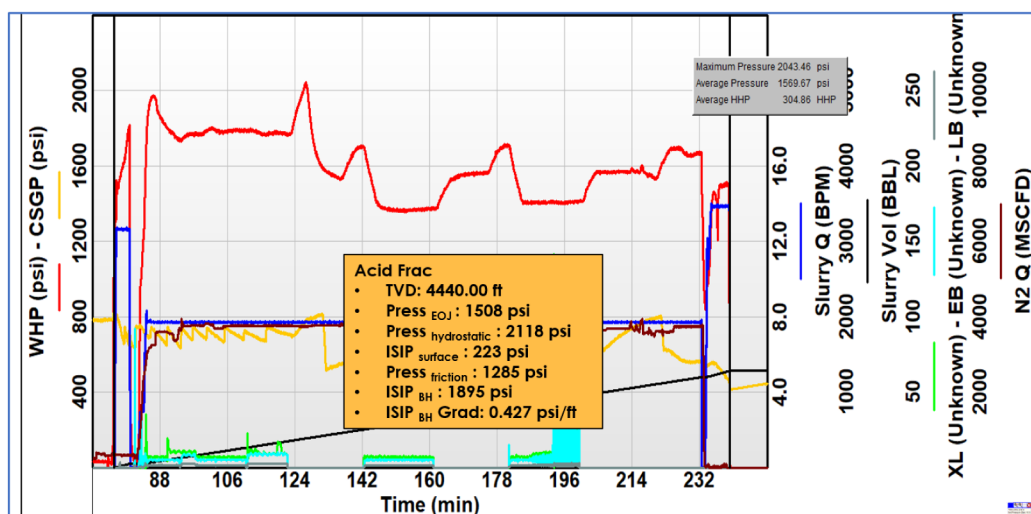


Figure 9—MainFrac Plot

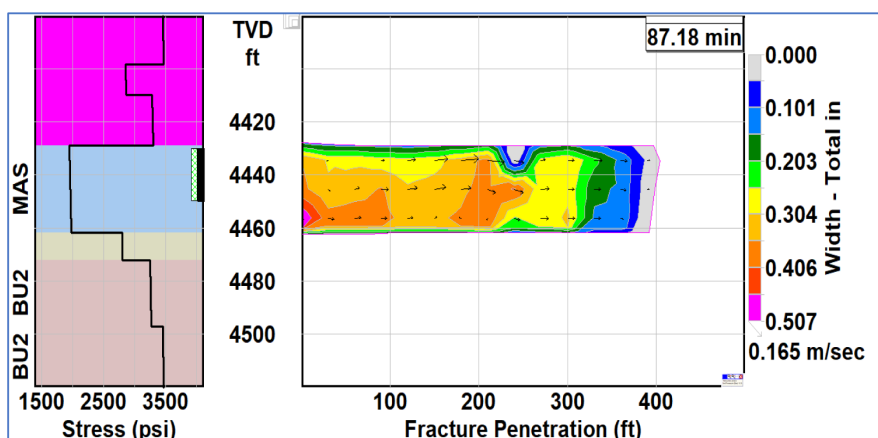


Figure 10—MainFrac Simulation Plot

Table 3—MainFrac Simulation Results

Hydraulic Fracturing Results		
Half Length	'Hydraulic' Length (ft)	367.5
	Acid Etched length (ft)	367.5
PRESSURE	Max Net Pressure (psi)	816.8
	Final Net Pressure (psi)	12.6
	Maximum Surface Pressure (psi)	2043.5
	Maximum Hydraulic Horsepower	601.0
	Average Acid Etched Width.(in)	0.33
HEIGHT:	Average Conductivity (md-ft)	7703.0
	Max Fracture Height (ft)	34.1
WIDTH:	Avg width at end of pumping (in)	0.33

Production Analysis & Conclusions

Based on a comparative performance analysis of offset wells, the application of energized fracturing fluids demonstrated a marked improvement in productivity compared to those stimulated with conventional non-energized fluids. Integration of temperature logs, bottom-hole pressure data, and sonic log interpretation confirmed that the induced fractures remained confined within the targeted reservoir layer, exhibiting an average fracture half-length of approximately 370 feet. Furthermore, the presence of pre-existing natural fracture networks was inferred from a notable increase in permeability, reaching values up to 1.2 Darcy, likely due to the enlargement of fracture apertures. Diagnostic parameters including Instantaneous Shut-In Pressure (ISIP), closure pressure, and time-dependent pressure decline trends collectively indicated the development of a well-connected, extensive fracture network propagating outward from the wellbore. Following stimulation, the well's production rate significantly increased six folds per day, demonstrating the effectiveness of the energized fracturing treatment.

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